The Great Lisbon Earthquake of 1755

by Gary Arndt

The Great Lisbon Earthquake
Everything Everywhere Daily
Play | See More
See how your data is managed...

Subscribe
Apple | Spotify | Amazon | Player.FM | TuneIn
Castbox | Podurama | Podcast Republic | RSS | Patreon

Podcast Transcript

On the morning of November 1, 1755, the citizens of Lisbon, Portugal set out to go to church for the feast of All Saints Day.

Little did they know that moments later, their lives and the lives of everyone in Lisbon were about to be turned upside down and that the city of Lisbon would almost cease to exist.

Learn more about the Great Lisbon Earthquake, one of the most devastating earthquakes in history, on this episode of Everything Everywhere Daily.

There have been many large earthquakes throughout history. Many of them share very common stories in terms of the devastation which they left behind.

What sets the Great Lisbon Earthquake apart from other such disasters is that it was really three different disasters that hit the city on the same day. All three of the disasters had the same genesis, but the sequence in which they hit made the day far more deadly than it would have been otherwise.

As I mentioned in the introduction, November 1 was a holy day on the Catholic calendar, which meant that everyone was supposed to attend church that morning. This is actually an important fact that will play into the story later.

The population of Lisbon in 1755 was approximately 275,000 to 300,000 people. Lisbon was the capital of the Portuguese empire which held territory in Europe, South America, Africa, Asia, and many smaller islands in between. This made Lisbon one of the most important cities in the world at that time.

At 9:30 in the morning, as people were going to or attending church, the first tremors struck. These were loud and noticeable, but they didn't cause much in the way of destruction.

The consul from the German city of Hamburg later reported, *"First we heard a rumble, like the noise of a carriage, it became louder and louder, until it was as loud as the loudest noise of a gun, immediately after that we felt the first tremble."*

Based on what happened, modern estimates are that there was a slip in the Azores–Gibraltar Transform Fault. This is a fault line that runs approximately from the strait of Gibraltar to the Azores Islands. One tectonic plate was thrust over another and the movement of land might have been as much as 1 kilometer.

The epicenter of the earthquake was believed to have been just off the southwesternmost point of Portugal near Cape St. Vincent.

At 9:40 am, the main earthquake hit. Church bells began ringing all over the city, and then just as quickly, buildings began to collapse. The churches, which were full of people, were the largest source of casualties.

For a period of 10 minutes, there were three major quakes that shook the city to its foundations. It is estimated that the earthquake measured 8.4 on the Richter scale.

To give you an idea of just how much the ground moved, in the middle of Lisbon's city center, there were fissures in the ground 5 meters, or 16 feet wide.

This was the first of the three disasters.

As the quake was in process, people rushed outside of their homes and buildings and ran to open areas where there was less chance of getting hit by falling debris.

The main open area was along the coast near the port. As people began crawling out of the rubble and assembling along the shore, they noticed something very strange happening.

The sea receded. The ocean which was right in front of them disappeared. What was now in front of them was a giant plain of mud. All of the sunken ships and debris which had been thrown in the sea over the centuries were visible.

What was happening was the beginning of the second disaster. A tsunami.

Approximately 40 minutes after the earthquake struck at around 10:30 am, the waterfront was struck by a massive 12 meter or 40-foot high wall of water. The water engulfed the port and the city center, flooding much of the collapsed ruins which had fallen less than an hour earlier.

People who were trapped in the rubble, shockingly enough, may have actually died from drowning.

Water flowed up the Tagus river so fast, that it kept pace with horses that were running at a full gallop.

After being hit by an earthquake and a tsunami in the space of an hour, the third disaster began to unfold.

In the homes and churches which weren't flooded with seawater, the candles which were burning, and the cooking fires which were ablaze, started fires that began to spread throughout the city.

Because everything was a pile of rubble at this point, little effort was made to put out the fire. It would have been near impossible to do even if they had wanted to given the difficulty of navigating streets filled with rubble.

The fire burned so hot, that it created a firestorm. It was so intense and consumed so much oxygen that people were killed from asphyxiation who was 30 meters away from the fire.

The fire burned out of control for five days.

In the end, 85% of the city had been destroyed, including all of the city center, the port, and most of the infrastructure involved in running the Portuguese Empire.

The royal palace was demolished by both the earthquake and the tsunami. The imperial library lost over 70,000 books. Many works of art by great European masters were lost forever.

The total number of deaths was estimated to be between 30,000 to 50,000 people.

The earthquake didn't just affect Lisbon, however. It also affected cities all along the coast of Portugal.

The city of Algarve was devastated as were many other smaller communities in the south.

There may have been as many as 10,000 people killed along the coast of Morocco.

The tsunami went across the Atlantic ocean causing damage to islands in the Caribbean and in Brazil.

The ramifications of the earthquake were wide-reaching and affected places around the world far removed from Portugal.

For starters, in the aftermath of the quake, there was a semi-organized relief effort. In the past, if something like this had happened, everyone would have mostly been on their own. Prime minister Sebastião de Melo organized the relief efforts including an organized corpse removal to avoid adding a fourth disaster to the city, disease. He did mass burials at sea to quickly and efficiently remove bodies, even over the objections of the Catholic Church.

He organized teams of firefighters once the initial firestorm died down.

The army was brought in to deter looting, and gallows were constructed in public areas around town to discourage looting.

On December 4, barely a month later, plans were unveiled for a new Lisbon. The new city of Lisbon was started from scratch with new wide streets. Unlike most European capitals, Lisbon is laid out like a modern, planned city for this reason.

The new buildings which were erected in Lisbon are known as Pombaline buildings named after the Marquês de Pombal, who was the chief city planner for the new Lisbon. Pombaline buildings were some of the first in the world to be built to resist seismic forces. They were also prefabricated, which allowed much of Lisbon to be rebuilt in record time.

The earthquake was also the beginning of the modern science of seismology or the study of earthquakes.

A survey was sent out to all of the parishes in Portugal asking them for information on how their location experienced the earthquake. They tried to systematically understand how the earthquake happened and how it propagated around the country.

The data collected in this survey is still located in the Portuguese archives and has been studied by modern scientists to try to recreate exactly what happened.

The other area affected by the earthquake was theology and philosophy.

This disaster occurred during the Enlightenment. There was a movement underway to try to understand the world through natural explanations, not only theological ones.

Normally, after a disaster, it would be explained that it was God's punishment for the sins of the people affected. There were some people in Lisbon who tried to argue that this was God's wrath, but they had a hard time explaining why all the Cathedrals were destroyed, but most of the brothels were spared.

Likewise, some Protestants in northern Europe tried to use the earthquake to prove that the Catholic church was not favored by God.

However, most scholars came to view this as simply a natural disaster. This is one of the first times a natural disaster was thought of as such, and it is why the 1755 earthquake has been called the world's first modern disaster.

I should note that while this was the worst earthquake to hit Lisbon and southern Portugal, it wasn't the first or only. There were other earthquakes in 1321, 1531, and 1969.

As with most cities in earthquake zones, it is likely there will be another major earthquake at some point in the future.

Next time, however, due to better construction, better planning, and better warning systems, Lisbon hopefully will never suffer the fate it did 226 years ago.

Everything Everywhere Daily, Podcast
‹ The Periodic Table of the Elements
› How Do Satellites Work?